
CS321M Project Pre-Analysis Plan

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1 Overview & Motivation

Large language models are increasingly proposed as automated evaluators—“judges”—in high-stakes tasks, including mental health screening. In this paradigm, a model receives a conversation transcript and produces a measurement-like judgment, such as symptom-item scores, a severity score, or a diagnostic classification. Although LLM-as-judge methods are increasingly used in clinical NLP, their sensitivity to interview structure has not been fully characterized. An under-examined assumption is that the judge attends to participant language rather than structural artifacts of the interview format. Previous work [Watawana et al., 2026] shows that trained classifiers can achieve high depression-detection F1 scores using only the interviewer’s scripted prompts, suggesting that models can exploit fixed prompt positions rather than genuine clinical cues.

Our main contributions are threefold: (1) an input ablation comparing participant-only, interviewer-only, and full-transcript conditions; (2) a reliability analysis of LLM-generated PHQ-8-style outputs, distinguishing absolute agreement from relative consistency across repeated calls and judge models, with a Generalizability Theory decomposition to identify which measurement facets drive rating variance; and (3) an exploratory gender-metadata sensitivity analysis in the participant-only condition, testing whether PHQ-8-style LLM judgments change when the same participant transcript is paired with different gender metadata. We hypothesize that zero-shot LLM judges will remain sensitive to interviewer-side artifacts and may also be sensitive to explicitly provided gender metadata.

This study aims to clarify whether LLM-based mental health assessment reflects participant symptom evidence or spurious conversational and demographic cues, and to motivate bias-aware evaluation protocols for clinical dialogue analysis.

2 Literature Review

Existing work on LLM-as-judge reliability has largely studied general-purpose NLP and dialogue evaluation rather than structured clinical interview settings. Zheng et al. [Zheng et al., 2023] introduce MT-Bench and Chatbot Arena, showing that strong LLM judges can approximate human preferences while also documenting judge-specific failure modes such as position bias, verbosity bias, and self-enhancement bias. Chiang and Lee [Chiang and Lee, 2023] further show that seemingly small choices in LLM-based automatic evaluation protocols can substantially affect agreement with human ratings. However, this line of work does not examine structural artifacts induced by semi-structured clinical interview scripts.

In parallel, automatic depression detection from clinical interviews has been studied as a promising approach for scalable and consistent mental-health assessment [Gratch et al., 2014, Valstar et al., 2016, Tao et al., 2023, Milintsevich et al., 2023]. DAIC-WOZ in particular has become a common benchmark for depression detection from multimodal clinical interviews [Gratch et al., 2014, Valstar et al., 2016]. Yet recent work raises concerns that models may exploit interviewer-side artifacts rather than participant symptom evidence. Burdisso et al. [Burdisso et al., 2024] show that DAIC-WOZ interviewer prompts can carry implicit diagnostic signal and that models using interviewer prompts may learn shortcut cues tied to specific scripted questions. Watawana et al. [Watawana et al.,

2026] extend this finding across DAIC-WOZ, E-DAIC, and ANDROIDS, showing that interviewer-only classifiers can match or outperform participant-only classifiers across multiple model families. Their analysis finds that interviewer-only models concentrate evidence on narrow scripted prompts, motivating bias-aware protocols that isolate participant turns.

Our project sits at the intersection of these two literatures. We ask whether zero-shot LLM judges exhibit the same prompt-induced bias previously observed in supervised clinical classifiers, and how such bias manifests in reliability and condition-sensitive measurement behavior. We study a clinical dialogue setting with known structural artifacts. We treat the LLM judge itself as a measurement instrument and evaluate it using psychometric tools, including ICC, Krippendorff’s α , and Generalizability Theory. This framing lets us ask not merely whether LLM judges predict PHQ-8 labels, but whether their ratings reliably measure participant symptom evidence or instead reflect spurious interviewer-script structure.

3 Proposed Methods & Analysis

3.1 Experimental Setup

Dataset. We use the DAIC-WOZ dataset, which includes 189 session folders and official train, development, and test split files. The split files provide participant IDs, gender, PHQ-8 total scores, and PHQ-8 binary depression labels. For the train and development splits, they also provide individual responses to each of the eight PHQ-8 questionnaire items. The PHQ-8 is a validated measure of current depression in the general population [Kroenke et al., 2009]. In this project, we use the development split.

Each interview is evaluated under three input conditions:

- **Participant-only (P):** only participant utterances are provided.
- **Interviewer-only (I):** only interviewer utterances are provided.
- **Full transcript (F):** both participant and interviewer utterances are provided.

LLM judges. Each transcript condition is presented to three LLM judges: GPT-4o, Claude 3.5 Sonnet, and Llama-3-70B. Each judge produces PHQ-8-inspired item-level evidence scores and a binary depression label. The judge also provides a short written rationale of two to three sentences. The rationale is used only for qualitative inspection of whether the judge cites participant symptom evidence or interviewer-script artifacts.

Repeated ratings. Each (interview, condition, judge) cell is rated three times at temperature = 0.3 to estimate intra-judge consistency.

3.2 Analysis 1 — Input Ablation

For the three input conditions—participant-only (P), interviewer-only (I), and full transcript (F)—we test whether transcript condition significantly affects the predicted total PHQ-8 score using a repeated-measures ANOVA. We additionally compute paired Cohen’s d_z effect sizes for the P–I, F–I, and P–F contrasts, with P–I treated as the primary contrast of interest. For binary depression labels, we compare the proportion of depression-positive predictions across conditions. For PHQ-8 item-level outputs, we examine whether interviewer-only inputs disproportionately increase scores on particular symptom items.

We hypothesize that I-condition outputs will differ systematically from P-condition outputs, indicating prompt-induced bias. In particular, if interviewer-only transcripts produce elevated PHQ-8 item scores or depression-positive labels, this would suggest that LLM judges may be relying on interview-script artifacts rather than participant symptom evidence.

3.3 Analysis 2 — Reliability and Generalizability Theory

Intra-judge reliability. For intra-judge reliability, repeated calls are treated as stochastic replicates of the same LLM judge under the same prompt and transcript condition. We compute ICC(2,1) to

estimate the reliability of a single randomly sampled call, and ICC(2,3) to estimate the reliability of the average of three calls. This quantifies whether observed differences between interviews are large relative to within-judge sampling variation.

Inter-judge agreement. For inter-judge agreement, we compare ratings across different LLM judge models within each input condition. For PHQ-8 item-level outputs, we compute Krippendorff’s α across judge models using an ordinal distance function, since each PHQ-8 item is scored on an ordered scale. For binary depression labels, we compute Krippendorff’s α with a nominal distance function, and optionally report pairwise Cohen’s κ as a secondary check. These metrics assess whether different LLM judges assign similar symptom-item scores and depression-positive labels to the same interviews after accounting for chance agreement.

G-theory variance decomposition. To decompose sources of measurement error, we fit a mixed-effects model using `pymr4`. For the summed PHQ-8-style severity score, we use the Gaussian mixed-effects model

$$Y \sim (1|I) + (1|C) + (1|M) + (1|I : C) + (1|I : M) + (1|C : M) + \epsilon.$$

where I denotes interview, C condition, and M judge model.

Here, repeated calls are treated as stochastic replicates and contribute to the residual variance. The corresponding variance components are

$$\sigma_{\text{interview}}^2, \sigma_{\text{condition}}^2, \sigma_{\text{model}}^2, \sigma_{\text{interview} \times \text{condition}}^2, \sigma_{\text{interview} \times \text{model}}^2, \sigma_{\text{condition} \times \text{model}}^2, \sigma_{\epsilon}^2.$$

We compute the G-coefficient $E\rho^2$ for relative decisions, such as ranking interviews by predicted depression severity, and the dependability coefficient Φ for absolute decisions, such as classifying an interview as depression-positive.

Bias diagnostic. The key diagnostic variance component is

$$\sigma_{\text{interview} \times \text{condition}}^2.$$

A large interview-by-condition interaction indicates that changing the transcript condition changes interview-level severity estimates in a non-uniform way. In this setting, it would mean that participant-only, interviewer-only, and full-transcript inputs do not preserve the same severity ordering of interviews. We interpret this as evidence consistent with condition-dependent measurement bias in LLM-as-judge ratings.

3.4 Analysis 3 — Gender Metadata Sensitivity in the Participant-Only Condition

As an exploratory fairness-oriented analysis, we examine whether LLM-as-judge outputs and reliability are sensitive to participant gender in the participant-only condition. We restrict this analysis to the P condition so that the comparison focuses on participant language rather than interviewer-side artifacts. The DAIC-WOZ split files include participant gender along with PHQ-8 total scores, binary labels, and PHQ-8 item-level responses, allowing us to study gender as a participant-level covariate.

We distinguish two related but conceptually different analyses. First, we conduct an observed-group stratification analysis using the gender metadata provided in DAIC-WOZ. In the standard P condition, gender is not explicitly provided to the LLM, although it may still be partially inferable from participant utterances. We stratify reliability by recorded participant gender, including repeated-call variability, inter-judge disagreement, PHQ-8 total-score error, and binary-label error rates. This analysis asks whether LLM-as-judge measurement is equally stable and valid across the gender groups represented in the dataset.

Second, we conduct a counterfactual gender-metadata sensitivity analysis within the P condition. For the same participant-only transcript, we create prompt variants that differ only in the gender metadata provided to the model: no gender provided, gender marked as female, and gender marked as male. This analysis does not estimate real-world gender fairness, because the participant transcript and true participant identity are held fixed while only the prompt metadata is manipulated. Instead, it tests whether the LLM judge is sensitive to explicitly supplied gender metadata when producing PHQ-8-style ratings.

We compare predicted PHQ-8 item-level scores and predicted binary labels across metadata variants. The main sensitivity statistic is the within-transcript difference in predicted total score:

$$\Delta_i = \widehat{\text{PHQ8}}_{i,\text{female}} - \widehat{\text{PHQ8}}_{i,\text{male}}.$$

If LLM judges are insensitive to provided gender metadata, these paired differences should be close to zero and binary-label flip rates should be low.

This analysis is inspired by the group-independence intuition in differential rater functioning: holding the participant-only transcript and PHQ-8 criterion variables fixed, the LLM’s score should not systematically depend on a sensitive attribute supplied in the prompt. Because the development split is small and the metadata intervention manipulates only the prompt rather than the participant or transcript content, we treat this analysis as exploratory and report uncertainty intervals rather than making strong causal or fairness claims.

4 Timeline & Responsibilities

We divide responsibilities between (1) data preparation and LLM-as-judge evaluation, and (2) reliability analysis and gender-sensitivity analysis.

Both contributors will jointly write the final report and interpret the measurement results.

Week	Milestone	Owner
Week 1	Clean P/I/F transcript files; remove non-semantic interviewer utterance identifiers; flag sessions with missing interviewer turns, interruptions, or other transcript-quality issues; merge PHQ-8 total scores, binary labels, gender, and eight item-level PHQ-8 responses into an analysis-ready label table; finalize the zero-shot judge prompt and pilot sanity-check outputs.	Both
Week 2	Complete the main LLM rating matrix. If all 35 development interviews pass the transcript-quality filter, this yields $35 \times 3 \times 3 \times 3 = 945$ structured rating records for the main P/I/F experiment.	Person A
Week 3	Complete input-ablation tables, item-level and total-score reliability tables by condition, binary-label agreement tables, variance-component table, $E\rho^2$ and Φ estimates plots comparing LLM outputs against self-reported PHQ-8 variables. Also run the participant-only gender-metadata sensitivity analysis using no-gender, female-gender, and male-gender prompt variants across the three LLM judges.	Person B
Week 4	Complete the final report, final figures, gender-sensitivity results, optional prompt-removal robustness results, and limitations/ethics discussion, including discussion of self-report labels, interviewer-script artifacts, gender-metadata sensitivity.	Both

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